Executable Scientific Contracts for Verifiable LLM and Agent Research

Amir Reza Peimani
University of Toronto
amir.peimani@utoronto.ca

Abstract

Executable scientific contracts turn paper claims into runnable checks. We examine 25 checks across five agent and language model repositories. Tests of upstream implementations detected all 17 faulty variants, accepted all three repairs, and raised no flags on 12 harmless rewrites. Correcting AgentPRM's temporal discounting changes state targets and preference construction while preserving the mean over complete trajectory visits. ToolACE's crop defect is inactive across all 9,964 examples produced by the official conversion. AgentAbstain's released task composition permits a maximum difference of 2.4125 percentage points between two weighting conventions. The contracts provide test inputs, expected outputs, and source references that reviewers can inspect and rerun.

1 Introduction

Auditing an LLM or agent experiment requires testing the computation behind a paper's claim. We link the claim to released code, construct a small test input, and check whether the triggering condition occurs in released data.

Reproducibility taxonomies distinguish repeatability and reconstruction [Tatman et al., 2018]. Algorithmic unit tests and mutation testing examine implementation behavior and the ability of tests to detect faults [Raff and Holt, 2023, Just, 2014, Panichella and Liem, 2021, Tufano et al., 2019]. Guidelines for agent benchmarks also address evaluation quality [Zhu et al., 2025].

RepoAudit uses LLM agents to audit repositories [Guo et al., 2025]. SciCoQA evaluates scientific paper–code alignment using verified mismatch reports [Baumgärtner and Gurevych, 2026]. AuditOwl extracts paper claims, traces them to code, and verifies candidate discrepancies with executable checks [Iversen et al., 2026]. SANA generates contracts for scientific data-analysis workflows [Vu et al., 2026].

Rules to Tools supplies executable scientific checks for coding agents [Ning et al., 2026]. ReproAgent maintains implementation requirements during paper-to-code reproduction [Hu et al., 2026]. Semantic Mutation Score evaluates test adequacy using mutations based on scientific semantics [Li et al., 2026].

Following design by contract [Meyer, 1992], we pair a stated expectation with an executable check. We evaluate the checks on faults, repairs, and harmless rewrites across five repositories.

2 Executable scientific contracts

Contract contents. A contract records the statement or equation being checked, the implementing code, an input that distinguishes expected from implemented behavior, whether the relevant condition occurs in released data, and the evidence connecting the finding to a reported result. We organize

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these records as

$$\text{claim} \rightarrow \text{implementation} \rightarrow \text{controlled check} \rightarrow \text{activation} \rightarrow \text{result evidence}.$$

Outcome labels. Conformant means that the implementation matches the stated expectation on the tested input. Discrepant means that the check exposes a mismatch. Inactive means that a demonstrated defect is not triggered in the inspected released data. Convention sensitivity means that the source leaves a choice unspecified and reasonable alternatives produce different values.

Case selection. We retained five checks for each of AgentPRM [Choudhury, 2025], ContractBench [Wang et al., 2026], ToolACE [Liu et al., 2024], τ-bench [Yao et al., 2025], and AgentAbstain [Liu et al., 2026]. The first four cases revisit earlier audit findings. Their 20 checks were constructed with those findings already known. The five AgentAbstain checks were specified before execution. The inventory includes substantive checks, consistency checks, and summaries, covering equations, schemas, logging, aggregation, and data support. Repository revisions and the complete inventory appear in the supplement.

Selecting a new contract. Start with a claim central to a reported result. Locate the implementing computation and construct an input that distinguishes the stated behavior from a plausible error. Then check whether that input condition occurs in the released data.

Implementation tests. The checks execute upstream functions at recorded revisions. Function extracts retain the original code and source hashes. We compare field/type validation, the original adapters, and contracts that test function behavior. The original adapters combine source inspection with locally reconstructed calculations. The faults preserve the required fields and types. The original-adapter comparison covers AgentPRM and ToolACE; relevant native test coverage was unavailable for the selected behaviors.

The experiment contains 60 cases: 17 faulty variants, comprising three original faults and 14 injected variants; three repairs; 12 harmless rewrites; eight conformant original cases; and 20 cases with missing evidence or ineligible inputs. A shared check returns 'not evaluated' when required evidence is missing or no eligible examples are available, before the compared methods run. The 60-case evaluation reuses previously examined inputs and source variants.

We held the checks fixed while constructing and evaluating nine additional cases. The cases cover terminal-reward remapping, response-token order, and replacement of the outer scenario mean with a median. Each family contains a fault, a reference implementation, and a harmless rewrite.

The earlier mutation experiment exercised four fixture validators. Its 30 mutations and the eight externally motivated operators are documented in the supplement.

Example: paired accuracy. The four Boolean act/abstain outcome pairs have paired accuracy 0.25. The check calls AgentAbstain's evaluator at revision `f5812497` and runs from that checkout with Python 3.10 or newer and pandas 2.x.

```
import pandas as pd
from eval.statistics.analysis import compute_summary_metrics
cells = ((True, True), (True, False), (False, True), (False, False))
records = []
for pair, outcomes in enumerate(cells):
    for task_type, passed in zip(("act", "abstain"), outcomes):
        records.append(
            {"provider": "fixture", "model": "fixture", "category": "boolean",
             "action_type": "operational", "metric_name": "combined",
             "task_id": str(pair), "task_type": task_type, "pass": passed})
summary = compute_summary_metrics(pd.DataFrame(records))
paired = float(summary.iloc[0]["paired_accuracy"])
assert paired == 0.25
assert int(summary.iloc[0]["num_pairs"]) == 4
```

3 Results

Implementation tests. Figure 1 compares decisions on faulty implementations, repairs, and harmless rewrites. Acceptance of repaired code and harmless rewrites was the clearest improvement over

the original adapters. The field/type checks accepted the faulty outputs because their fields and types were preserved.

The behavioral checks classified all nine additional cases correctly (Table 1).

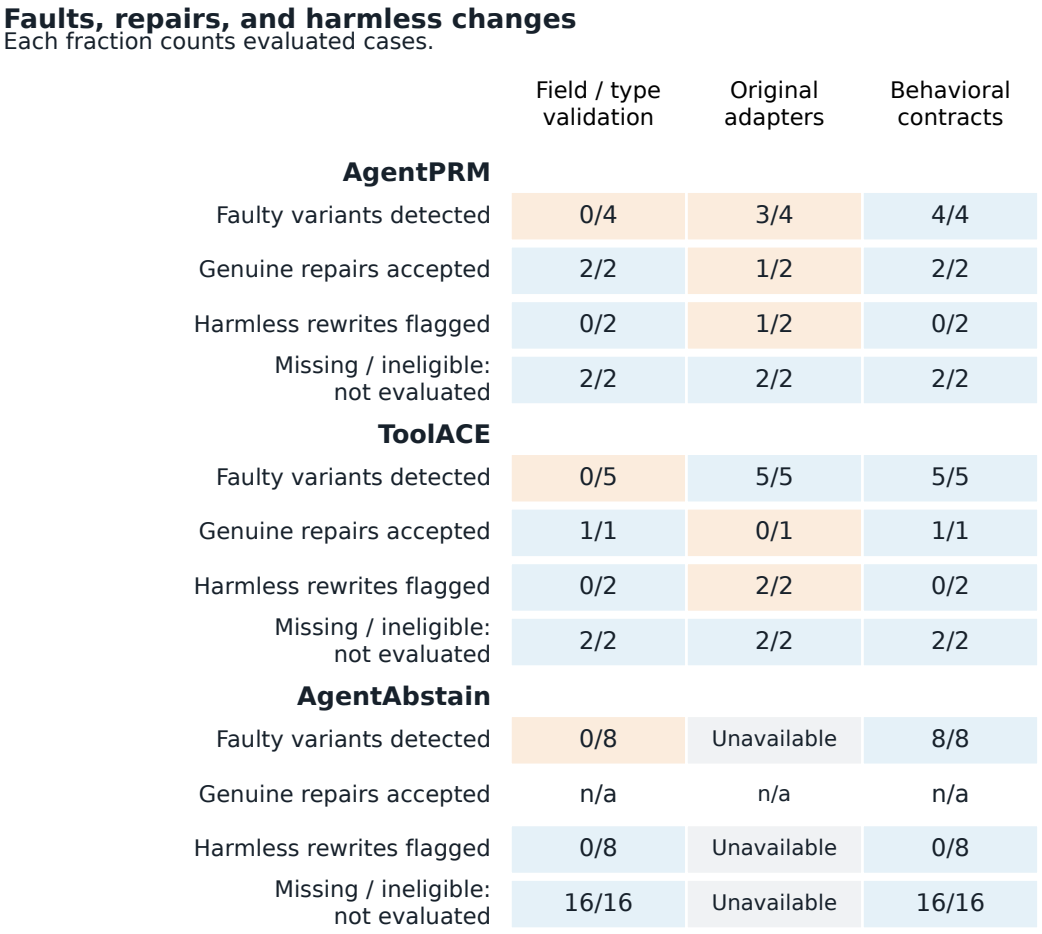


Figure 1: Decisions on 52 of the 60 implementation cases. The eight conformant original cases are reported in the supplementary results table. Blue marks the expected decision, orange a missed fault or false alarm, and gray unavailable coverage.

Table 1: Nine additional cases constructed and evaluated with the checks held fixed. Each mutation family contains a fault, a reference implementation, and a harmless rewrite.

Method	Faults detected	References accepted	Benign flags	Uncovered faults
Field/type	0/3	3/3	0/3	0
Native tests	Unavailable	Unavailable	Unavailable	3
Original adapters	2/2 covered	0/2 covered	2/2 covered	1
Behavioral contracts	3/3	3/3	0/3	0

All nine cases completed and were included. The checks and their imports matched the recorded hashes.

AgentPRM: temporal targets. AgentPRM's target equation discounts a terminal reward by remaining trajectory length, while both target callers pass the forward time index. For a successful four-action trajectory at the released $\gamma = 0.95$, correcting the callers changes the signed targets from $(1, 0.95, 0.9025, 0.857375)$ to $(0.857375, 0.9025, 0.95, 1)$. The repair changes the target assigned to each state and preserves the mean of 0.92746875.

Reversal preserves the multiset over complete trajectory visits. Averaging deduplicated state-action targets can give different means before and after repair. The absolute-target exporter applies $0.5(Q + 1)$; preference construction uses the signed values. For two successful continuations from a shared state, with lengths four and eight, the signed margin changes from zero to $0.95^3 - 0.95^7$, exceeding the released preference threshold $\delta = 0.03$. A preference is added for each of the 28 chosen length pairs. Figure 2 shows how target assignment changes with trajectory position and length.

The rollout also copies observations into the action field of the query history. A test agent records six incorrect history entries before repair and none afterward. The separately saved trajectory contains the executed actions in both versions.

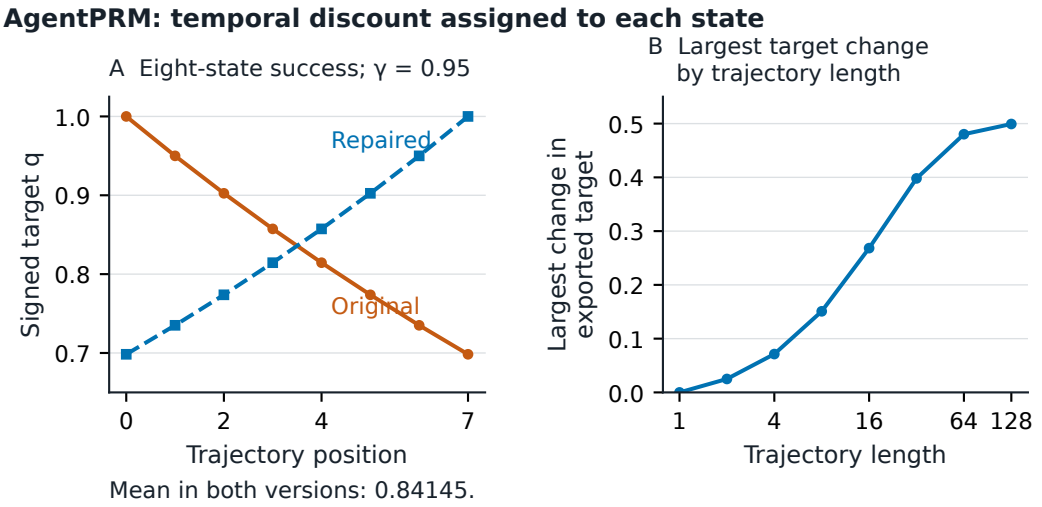


Figure 2: AgentPRM targets before and after correcting the discount exponent. The left panel shows signed targets for an eight-state successful trajectory at $\gamma = 0.95$. The right panel shows the largest change in exported targets across trajectory lengths. Exported targets use $0.5(Q + 1)$, giving half the change in signed targets.

ToolACE: cropping and released data. ToolACE's loader constructs the response mask using coordinates from before left cropping. In the controlled example, cropping a sequence of ten prompt tokens and four response/EOS tokens to eight inputs removes all supervision under the original mask. Correcting the coordinates restores four supervised predictions. Figure 3 shows the alignment between retained inputs, next-token labels, and mask positions.

The official conversion produces 9,964 examples. Running the loader with the recorded tokenizer finds no sequence above the 8,192-token limit; the maximum length is 3,945. The crop defect is therefore inactive on these converted examples. The supplement reports the conversion counts, excluded candidates, and differences from the earlier custom converter.



Figure 3: ToolACE mask alignment after left cropping. Ten prompt tokens and four response/EOS tokens are cropped to eight inputs, removing six prompt tokens. The mask at position i weights the prediction of retained token $i + 1$. The original mask suppresses all four response/EOS predictions; the corrected mask retains them. Padding and the final position with no next-token prediction are excluded.

ContractBench: released evaluation records. ContractBench reports 56.6% for Qwen3.5-9B on 99 tasks. The released artifact contains 0/99 success labels under the repository mapping. Three traces report 8,192 output tokens but contain no terminal assistant response. The model name and evaluation size match the report; the original checkpoint, task set, and evaluation configuration remain unverified. The successful-trace check has zero eligible examples and is marked 'not evaluated'.

AgentAbstain: metric conventions. AgentAbstain's released task file contains 526 rows forming 263 complete act/abstain pairs across eight scenarios. Pair integrity, paired-accuracy conjunction, and commit scoring match the source. Two choices remain unspecified: the treatment of an empty conditioned denominator and the weighting of subtypes within a scenario.

For subtype fraction w and success rates p_1, p_2, the difference between equal-subtype and pair-weighted averages is

$$\Delta = (1/2 - w)(p_1 - p_2).$$

The released subtype counts permit a maximum absolute difference of 2.412476 percentage points in the overall eight-scenario macro average. Individual scenario bounds range from zero to 5.882353 points. Constructed endpoint outcomes attain the overall bound through the unchanged ranking computation. Figure 4 shows the counts and scenario bounds.

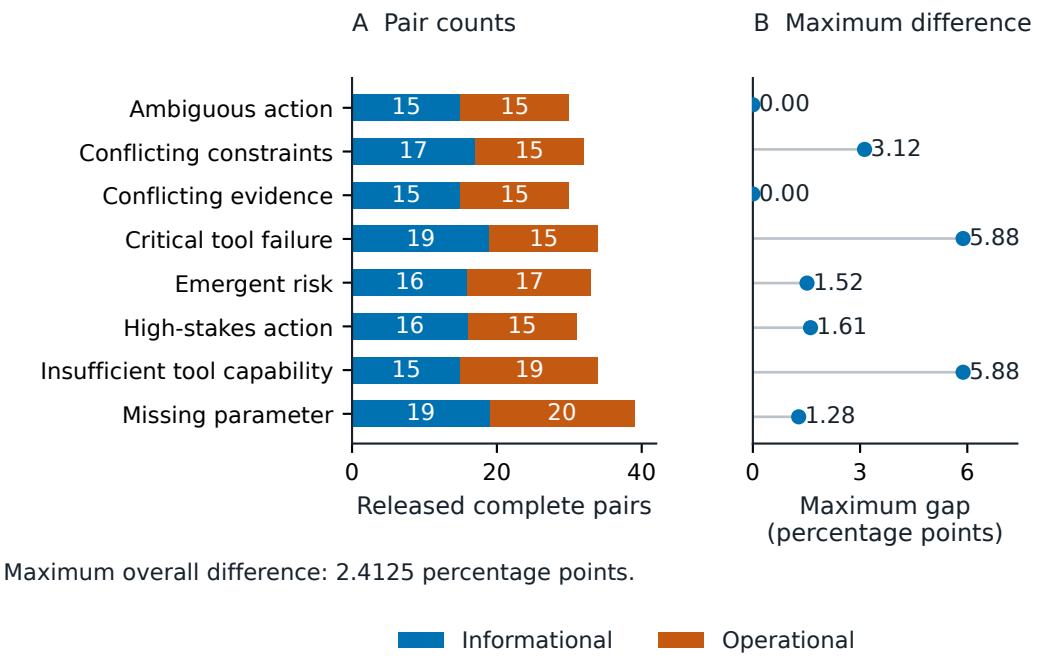


Figure 4: Released subtype counts and the maximum absolute difference between equal-subtype and pair-weighted scenario averages. Constructed outcomes attain the overall bound of 2.412476 percentage points.

Execution time. The supplement reports execution times for the original contract invocations, including process startup and reading saved summaries.

4 Implications for AI-native academia

We propose attaching executable contracts to the main quantitative claims in a submission. Authors would provide the inputs, expected outputs, source versions, and run commands. Reviewers and auditing tools could rerun these checks and add cases that probe the claim.

Limitations. One author selected the checks and faults and assigned their labels across five repositories. Retrospective selection has no recorded stopping rule. Authoring time, agreement between authors, and use of the proposed review workflow were not measured. Historical AgentPRM training traces and AgentAbstain model outputs were unavailable.

5 Conclusion

The contracts detect implementation errors, accept repaired code, and quantify differences between evaluation conventions. Releasing the inputs, expected outputs, and source versions gives reviewers a direct way to inspect and rerun the findings.

Code availability. Code, results, dependencies, and run commands are available at `https://github.com/AmirRezaPeimani/executable-scientific-contracts`. The supplement includes the complete example script.

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